

Bharatiya Antariksh Hackathon 2026



ip3a

Problem Statement: *Detection and Characterization of Subsurface Ice in Lunar South Polar Regions Using Chandrayaan-2 Radar and Imagery Data for Landing Site and Rover Traverse Planning*

Can you discover hidden lunar ice deposits and design India's future robotic exploration strategy around them?

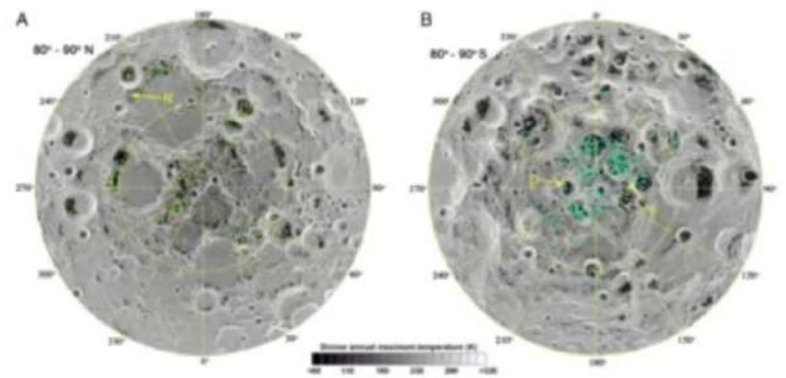
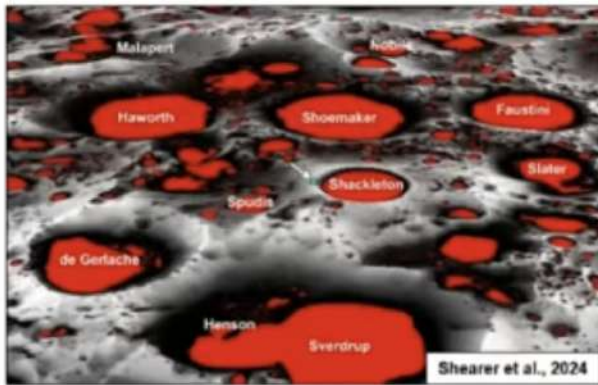
I have Chandrayaan-2 radar and imagery data. I need to find subsurface ice, select a landing site, plan a rover route, and estimate ice volume.

DFSAR + OHRC Data
↓
Detect Subsurface Ice
↓
Select Landing Site
↓
Design Rover Traverse
↓
Estimate Ice Volume (0–5 m)

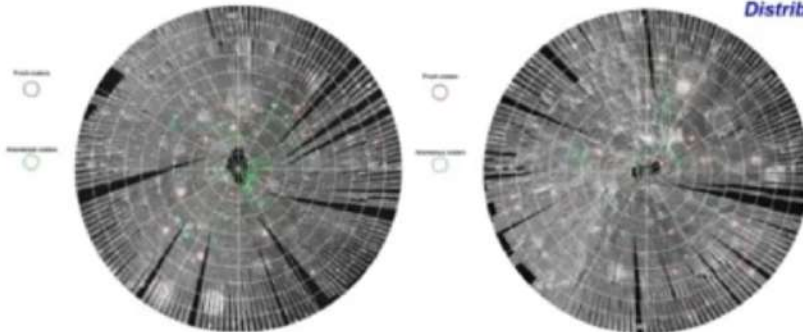
What students should feel proud of:

- *Where exactly is subsurface ice located?*
- *How can ice be distinguished from rocky terrain?*
- *Which sites are safe for landing?*
- *How can a rover reach the target efficiently?*
- *How much ice may actually exist?*

Evidence for Lunar Polar Ice — But Are We Done Yet?



Distribution of ice-bearing pixels constrained by M3, LOLA and LAMP (Li et al., 2018)

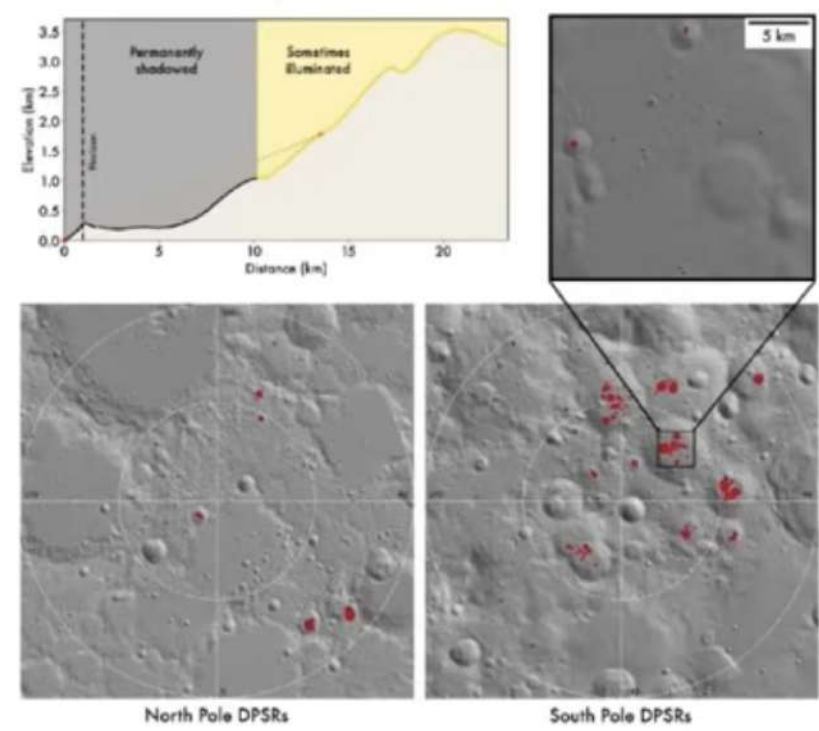


Location maps for the anomalous craters in north and south polar regions that have higher than average CPRs (modified from Spudis et al., 2013)

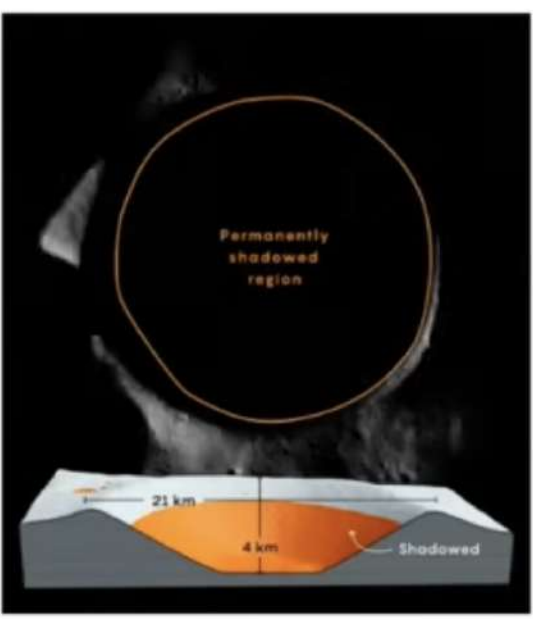
Key gap areas in polar ice detection:

- Ambiguity in radar-based detection
- Poor constraints on ice depth and volume
- Incomplete understanding of micro environments
- Ice formation and delivery mechanisms
- Surface vs. subsurface discrimination
- Spatial heterogeneity

What are 'Doubly Shadowed' craters?



A point is doubly shadowed if it is permanently shadowed and has no direct line of sight to any non-permanently shadowed surfaces.



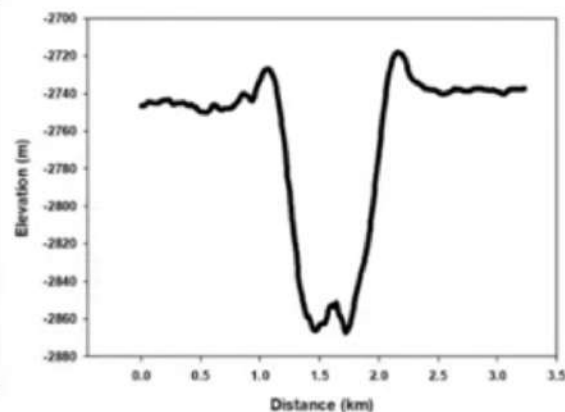
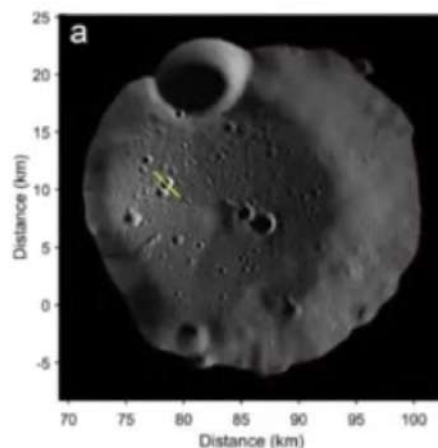
The north-south axis of the Moon is nearly perpendicular to the plane of the solar system. The rims of craters block sunlight from reaching their depths.

Merrill Sherman/Quanta Magazine; Source: NASA

The rim of the doubly shadowed crater shields its interior from scattered light and thermal emission from the periodically illuminated rim of the larger crater. These double permanently shadowed regions (DPSRs) are among the coldest places in the inner solar system, with theoretical temperatures approaching 25 K—the minimum possible surface temperature on the Moon.

Table 1
Size and Location of the Largest Doubly Shadowed Regions on the Moon

Area (km ²)	D ₅₀ (m)	Latitude (°)	Longitude (°)	Lunar Features
0.274	590.219	85.552	49.426	Filipier, Byrd D, Nansen F Crater
0.262	577.464	~88.275	38.992	Shoemaker Crater
0.242	555.204	~85.362	48.430	Noble Crater
0.220	529.858	85.636	50.220	Filipier, Byrd D, Nansen F Crater
0.213	521.135	~86.870	24.335	Haworth, Shoemaker Crater
0.201	505.509	~87.390	82.312	Faustini Crater
0.187	488.211	86.035	36.598	Filipier Crater
0.182	481.118	~86.940	19.223	Haworth, Shoemaker Crater
0.176	473.919	~87.114	85.381	Faustini Crater
0.167	461.671	86.113	36.769	Filipier Crater
0.152	440.068	~88.171	117.626	Slater Crater
0.126	400.535	~87.915	~10.942	Haworth Crater
0.121	391.858	~87.381	~6.105	Haworth Crater
0.116	384.478	87.086	158.568	Bosch Crater
0.114	381.486	~87.841	40.078	Shoemaker Crater
0.111	375.430	~87.436	80.655	Faustini Crater
0.076	312.094	~87.310	86.333	Faustini Crater
0.071	300.878	~88.531	~141.567	Sverdrup Crater
0.070	298.967	~88.049	41.710	Shoemaker Crater
0.068	295.109	89.598	~54.699	Hinschwood Crater
0.063	283.221	~89.053	~45.939	de Gerlach, Haworth Crater
0.061	279.146	~87.138	~8.665	Haworth Crater
0.058	270.811	87.621	151.865	Bosch Crater



An example of a doubly shadowed crater within Faustini PSR

Expected Outcomes

- Identification of *high-probability subsurface ice regions* in doubly shadowed craters
- A *validated radar-based detection framework* for subsurface ice
- A feasible *landing site* near scientifically relevant targets
- An optimized *rover traverse path*
- Quantitative estimates of *subsurface ice volume*

Suggested Tools/Technologies

- GIS platforms: QGIS / ArcGIS
- Programming: Python (NumPy, SciPy, GDAL, rasterio)
- Image processing: ENVI
- **DFSAR Processing: MIDAS**
- Terrain analysis: Digital Elevation Models (DEM tools)
- Path planning: Optimization algorithms / AI-based navigation models
- Visualization: QGIS / ArcGIS / MATLAB / Python-based plotting

Expected Solution / Steps to be followed to achieve the objectives

- Map permanently shadowed regions and identify **doubly shadowed craters** using illumination models and OHRC imagery.
- Analyze **DFSAR data** to compute radar parameters such as Circular Polarization Ratio (CPR) and Degree of Polarization (DOP). Apply refined criteria (e.g., $CPR > 1$ and $DOP < 0.13$) to identify potential subsurface ice signatures.
- Study crater morphology, slopes, boulder distribution, and surface roughness using **OHRC data**.
- Evaluate terrain safety and proximity to **ice-bearing regions**.
- Design an **optimal and safe path** considering terrain hazards and solar power constraints.
- Use radar backscatter models and dielectric assumptions to estimate ice concentration and volume within the top 5 meters.

In this Challenge #8, we are asking you to think like mission planners. Your task is to move from orbital observations to an actionable exploration strategy: where to land, where to drive, where to excavate, and how much ice may be available.

Your challenge is to discover hidden lunar ice and design a realistic pathway for future ISRO missions to reach and utilize it.